Technical Document #736: Cybersecurity - Comprehensive Overview

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Incident response plans define how organizations respond to security incidents. They include preparation, detection, containment, and recovery phases.

Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Network segmentation divides networks into smaller segments. It limits the spread of attacks and improves security monitoring.

Social engineering manipulates people into revealing confidential information. Phishing is the most common social engineering attack.

Security information and event management (SIEM) systems collect and analyze security logs. They provide real-time visibility into security events.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Multi-factor authentication (MFA) requires multiple forms of verification.
- Encryption converts plaintext into ciphertext to protect data confidentiality.
- Social engineering manipulates people into revealing confidential information.